

Human Development Index (HDI) Predictor

Coding & Solution

Project: Human Development Index (HDI) Predictor

Phase: Project Development Phase

1. Dataset Generation

generate_dataset.py builds a 144-country dataset using realistic Life Expectancy, Mean Years of Schooling, GNI per Capita, and Internet Users values, then computes HDI using the official UNDP geometric-mean formula (health index $\times$ education index $\times$ income index) with a small amount of noise added for realism.

2. Model Training

train_model.py performs EDA (a strip plot of schooling vs HDI, and a correlation heatmap), fills missing values with column means, label-encodes the Country feature, splits the data 90/10, fits a Scikit-learn LinearRegression model, evaluates it with R^2 , MAE, and RMSE, and pickles the model and encoder into the Flask/ folder.

3. Flask Backend

app.py loads HDI.pkl and country_encoder.pkl at startup and exposes three routes: / and /Home render the landing page; /Prediction renders the input form pre-populated with known countries; /predict (POST) encodes the submitted country, builds a feature DataFrame, predicts the HDI score, clamps it to a valid [0, 1] range, classifies it into a development tier, and renders the result page.

4. Error Handling

The /predict route is wrapped in a try/except block so invalid input (e.g. an unrecognized country or non-numeric value) returns a friendly error message on the result page instead of crashing the server.